Baselines and fair comparisons

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Ask whether the comparison changes method, information, labels or resources.

Teach fair baselines without treating missing conditions or toy scores as real benchmark assurance.

Keep one habit: Model source confound → align permitted evidence → protect hidden evaluation labels → assess task budget → retain unreported resources and inference limits.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of **baseline**s and fair comparisons

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Hypotheses and measurable outcomes. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of hypotheses and measurable outcomes

Reference example, not the only acceptable wording: Teacher explanation: A hypothesis predicts a measurable comparison under stated conditions. Declare primary outcome and direction before inspecting results.

Toy hypothesis:method B improves supported-answer rate over **baseline** A on a reserved paired set. Define support rubric and failure treatment; record secondary outcomes separately.

Without opening the old lesson, explain one key idea from Replication and reproduction. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of replication and reproduction

Reference example, not the only acceptable wording: Teacher explanation: Reproducing a computation uses recorded artifacts; replication collects new **evidence** under a comparable question.

Naming conventions vary, so define the terms in the project. Here replay means same artifacts/procedure; new-data replication means newly collected eligible cases under the same plan.

Preserve environment, data versions and raw results.

Without opening the old lesson, explain one key idea from Reading a research claim. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of reading a research claim

Reference example, not the only acceptable wording: Teacher explanation: Reading a claim requires its method, data, comparison and uncertainty. Titles and abstracts may omit conditions crucial to interpretation.

Separate the claim from supplied **evidence**. Record dataset, unit, **baseline**, metric and limitations. Ask whether the statement extends beyond those **observation**s.

Pictured end goal

Explain **baseline**s and fair comparisons using the records and boundaries of this fictional investigation.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Retrieval answer guidance

Hypotheses and measurable outcomes

Replication and reproduction

Reading a research claim

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. Fair **baseline**s use credible procedures on matched **evidence** and resource conditions. A weak arbitrary reference can exaggerate improvement.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of **baseline**s and fair comparisons

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up
</details>

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach a credible baseline with matched evidence and explicit constraints

> Separate a source confound, test leakage and unmeasured resource trade-offs.

Definitions and complete fictional comparison contract

Fair **baseline**s use credible procedures on matched **evidence** and resource conditions. A weak arbitrary reference can exaggerate improvement.

For a room-reporting study use manual authoritative lookup or a deterministic lookup matching the requested room, not random room names. Match cases, available sources and scoring.

Fair-comparison item	Complete fictional classroom rule
Credible baseline	Manual authoritative lookup or deterministic lookup matching the requested room; not arbitrary random rooms
Matched evidence	Same eligible current sources for a method-focused comparison
Matched workload/scoring	Same fixed case IDs, outcome rubric and evaluation labels kept out of tuning inputs
Resource policy	Declare timing boundary, budgets and what each procedure actually consumes; unknown is not automatically matched
Added main fixture	Same lookup function on clubs V/W/X; only source freshness differs

Fair-comparison item	Complete fictional classroom rule
Current register/truth	V→E, W→B, X→D
Old register	V→C, W→B, X→A
Separate picture	Same eight fixed questions and allowed sources; A 6/8 correct, B 7/8; no test-case tuning
Limits	Constructed counterexample/aggregates, not a model benchmark, measured runtime, stable advantage or significance result

Main counterexample ledger and checked reasoning

Added same-procedure fixture	Current correct room	Old-source lookup	Current-source lookup
Club V	E	C: incorrect	E: correct
Club W	B	B: correct	B: correct
Club X	D	A: incorrect	D: correct
Score on three constructed cases	Same judge and function	1/3	3/3
Align both to current register	Same cases/function/current evidence	3/3	3/3
Meaning	Only information changed in this demonstration	Score gap can occur without a method difference	Not a prediction that real A/B models must tie

Main unequal register freshness prevents isolating method improvement. Added identical lookup fixture scores 1/3 old versus 3/3 current and ties 3/3 when sources aligned; not a claim that real models tie. Picture A 6/8=75%, B 7/8=87.5%, net one correct/+12.5 points; raw paired flips and resource data absent.

Hidden judge-key exposure invalidates unseen testing, while permitted current task register is legitimate evidence. Small aggregates alone do not supply a significance result; do not teach that small N makes all significance impossible.

Selected model 12 seconds versus baseline 2 with per-case budget 3: model exceeds budget even if quality rises; compare complete procedures under declared budgets or matched time limits, not invented early answers.

Baselines and fair comparisons — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Assistant gets current register; baseline gets an obsolete one. Assistant scores higher.	For a room-reporting study use manual authoritative lookup or a deterministic lookup matching the requested room, not random room names.	Source freshness is confounded with procedure. Provide both the same eligible current sources or report the unequal comparison without isolating method.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Source freshness is confounded with procedure. Provide both the same eligible current sources or report the unequal comparison without isolating method.

Qualification: A **baseline** is credible relative to the task, not merely a procedure easy to outperform.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Assistant gets current register; **baseline** gets an obsolete one. Assistant scores higher.

Source freshness is confounded with procedure. Provide both the same eligible current sources or report the unequal comparison without isolating method.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Credibility, leakage and budget probes

Fair-baseline probe	Expected interpretation
Baseline intentionally random despite available credible lookup	Weak reference can exaggerate apparent improvement
B current sources, A obsolete sources	Information freshness confounded with method
Both use current permitted register	Appropriate matched task evidence; not automatically all other controls
Only B sees hidden judge labels	Evaluation leakage; not an unseen fair test
Same headline score but resources unknown	Cannot infer equivalent cost or budget performance
Higher quality violates task time budget	Report complete procedure trade-off; no budget-feasible superiority claim
Small aggregate score gap	Descriptive result; stable advantage/significance need appropriate evidence
Unknown paired case outcomes	Net correct-count change does not identify every gain/loss
Conditions audit has no recorded mismatch	Checks declared metadata only, not actual execution verification

Separate eight-case gain and missing-condition scope

A **baseline** deserves a fair chance. Two classifiers answer the same eight fixed questions with the same allowed sources.

Simple rule A: 6/8 correct. New model B: 7/8 correct. No test case is used to tune either version.

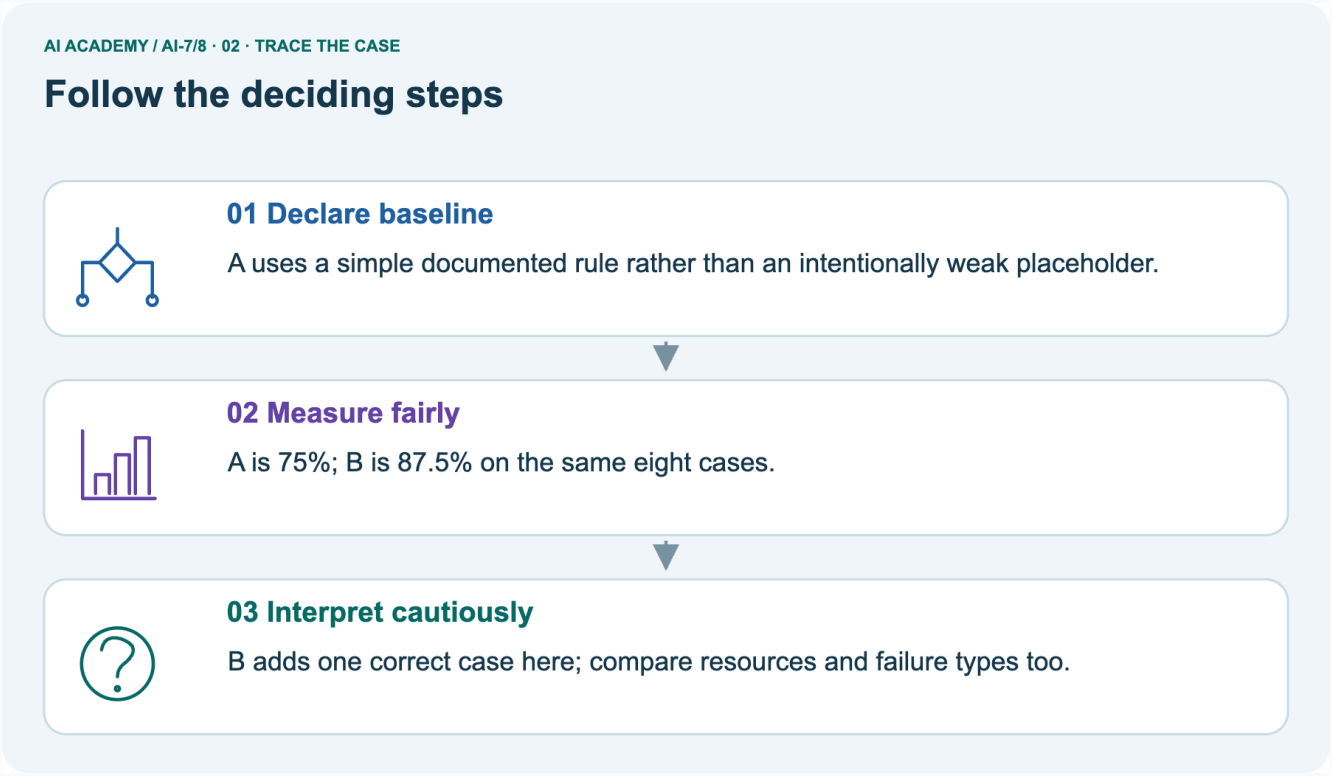
The picture is a separate eight-case baseline/model comparison. A 6/8 =75%; B 7/8 =87.5%; difference 12.5 percentage points and one net additional correct answer. Aggregates do not reveal every paired gain/loss. Cases, allowed sources and scoring are matched in the

scenario, with no test-case tuning; resources remain unreported. Giving B the hidden evaluation answer key would break the unseen comparison.

An approved source register that the task permits reading is not the same as exposing held-out judge labels. The supplied aggregate comparison has no statistical analysis or broad-advantage evidence, and should not be reported as one.

Separate eight-case picture	A baseline	B new model	Interpretation
Correct/total	6/8	7/8	Same fixed case denominator
Observed rate	75%	87.5%	B higher on supplied cases
Difference	Reference	+12.5 percentage points	One net additional correct count; paired flips not supplied
Permitted sources/cases	Same declared sources/cases	Same declared sources/cases	Relevant matching conditions
Tuning on test answers	None in original scenario	None in original scenario	Changed answer-key exposure breaks unseen evaluation
Resource/timing evidence	Not supplied	Not supplied	Cannot infer budget eligibility or efficiency

Picture trace and exposed-key key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Declare baseline	A uses a simple documented rule rather than an intentionally weak placeholder.
2. Measure fairly	A is 75%; B is 87.5% on the same eight cases.
3. Interpret cautiously	B adds one correct case here; compare resources and failure types too.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **baseline** anchors the claim of improvement. Fair comparison keeps information access, cases and scoring aligned while allowing the declared method difference.

These eight-case aggregate results alone establish neither a stable advantage nor a reported statistical-significance result.

Change: Give B the answer key before testing.

Prediction: The resulting comparison is invalid as unseen evaluation.

Reason: B would receive privileged information that A did not receive.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Baseline	A relevant reference procedure
Matched comparison	A comparison with aligned task conditions
Resource budget	The permitted time or computation

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

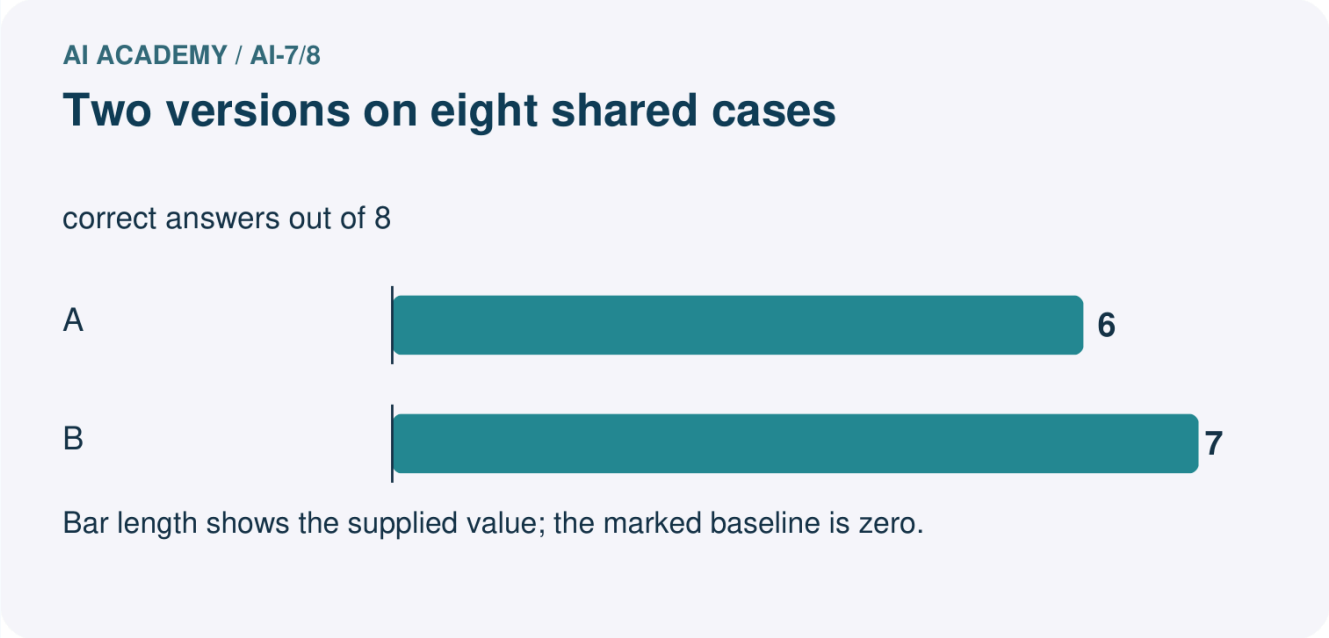
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Baselines and fair comparisons

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/8

Baselines and fair comparisons

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Declare baseline

A uses a simple documented rule rather than an intentionally weak placeholder.



2. Measure fairly

A is 75%; B is 87.5% on the same eight cases.



3. Interpret cautiously

B adds one correct case here; compare resources and failure types too.

CHANGE ONE THING

Give B the answer key before testing.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Assistant gets current register; **baseline** gets an obsolete one. Assistant scores higher. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Source freshness is confounded with procedure. Provide both the same eligible current sources or report the unequal comparison without isolating method.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Baseline**s and fair comparisons. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A **baseline** is credible relative to the task, not merely a procedure easy to outperform.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Calculate 6/8 and 7/8 correctness and name the one-case gain. Why is giving B the answer key incompatible with an unseen comparison to the simple **baseline**?

A uses a simple documented rule rather than an intentionally weak placeholder. A is 75%; B is 87.5% on the same eight cases. B adds one correct case here; compare resources and failure types too.

Changed case: The resulting comparison is invalid as unseen evaluation. B would receive privileged information that A did not receive.

Changed-case reasoning: The resulting comparison is invalid as unseen evaluation. B would receive privileged information that A did not receive.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main current/obsolete-source comparison confounds freshness with procedure. Added same-function fixture illustrates that confound without measuring a model. Picture75/87.5% netgain is descriptive, with unseen labels protected and resource evidence absent. Selected model 12 seconds per case exceeds the 3-second task budget, while baseline 2 seconds fits; quality gain alone cannot imply budget-feasible improvement.

Compare complete procedures under declared budgets or matched time limits and keep all outcomes/resource measurements visible.

Diagnose the current misconceptions

The worked result establishes every future case: A **baseline** is credible relative to the task, not merely a procedure easy to outperform.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Model uses 12 seconds and **baseline** 2 on each case. Task has a 3-second budget. How should results be framed?

Teacher explanation: The model exceeds the task budget even if quality rises. Compare complete procedures under declared budgets or use matched time limits.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Fair **baseline**s use credible procedures on matched **evidence** and resource conditions. A weak arbitrary reference can exaggerate improvement. For a room-reporting study use manual authoritative lookup or a deterministic lookup matching the requested room, not random room names. Match cases, available sources and scoring.

W2 Trace the worked case

Assistant gets current register; **baseline** gets an obsolete one. Assistant scores higher. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Source freshness is confounded with procedure. Provide both the same eligible current sources or report the unequal comparison without isolating method.

W3 Audit the limit

A report omits this qualification: A **baseline** is credible relative to the task, not merely a procedure easy to outperform. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A **baseline** is credible relative to the task, not merely a procedure easy to outperform.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A **baseline** is credible relative to the task, not merely a procedure easy to outperform.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: Source freshness is confounded with procedure. Provide both the same eligible current sources or report the unequal comparison without isolating method.

Qualification: A **baseline** is credible relative to the task, not merely a procedure easy to outperform.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Model uses 12 seconds and **baseline** 2 on each case. Task has a 3-second budget. How should results be framed?

Teacher explanation: The model exceeds the task budget even if quality rises. Compare complete procedures under declared budgets or use matched time limits.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: The model exceeds the task budget even if quality rises. Compare complete procedures under declared budgets or use matched time limits. Require intermediate reasoning and the stated scope.

Fresh retry: Candidate receives full source, **baseline** only title. Expected reasoning: **Evidence** availability differs. Equalise access before attributing gains to the candidate method.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

Add a labelled **baseline**s and fair comparisons **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **baseline**s and fair comparisons.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Experimental units and independence. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/9, ai-7/11, ai-7/14. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of **baseline**s and fair comparisons

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: Assistant gets current register; **baseline** gets an obsolete one. Assistant scores higher. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Source freshness is confounded with procedure. Provide both the same eligible current sources or report the unequal comparison without isolating method.

Check every learner before discussion: Use the worked case for **Baseline**s and fair comparisons. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A **baseline** is credible relative to the task, not merely a procedure easy to outperform.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Give B the answer key before testing. Require a **prediction** and a reason before comparing. Expected result: The resulting comparison is invalid as unseen evaluation.

Reason: B would receive privileged information that A did not receive.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

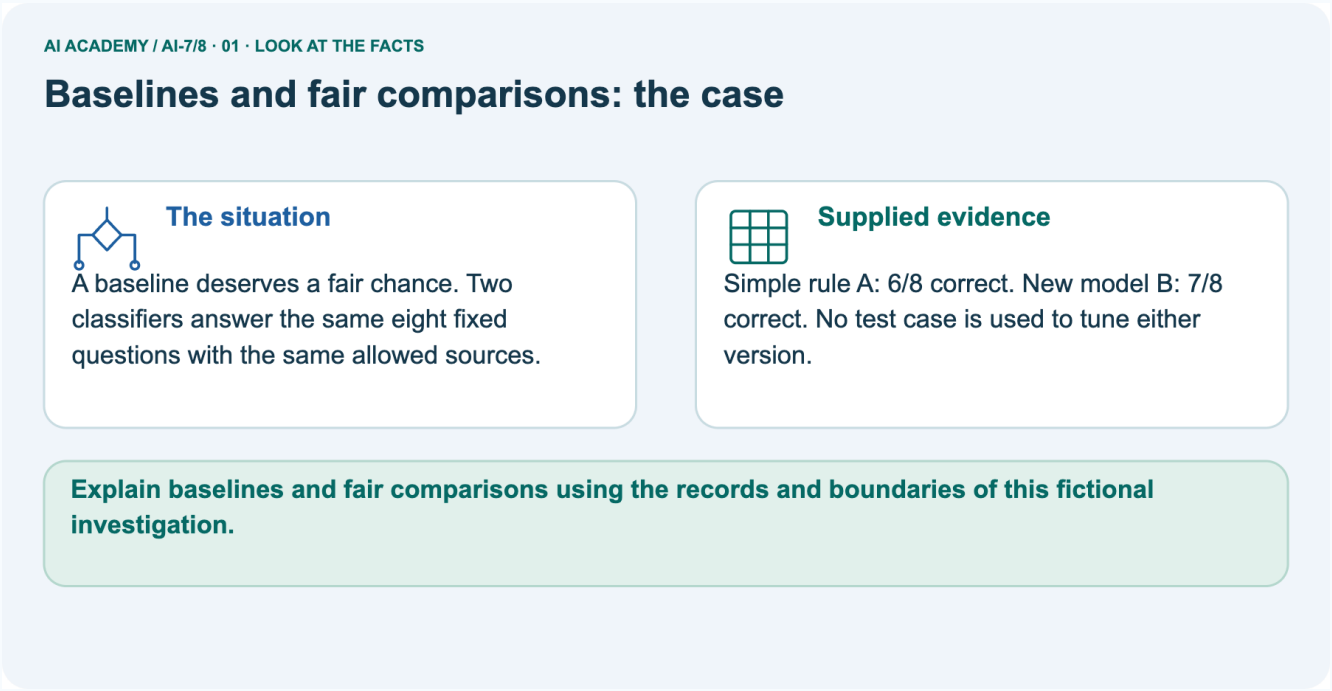
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Baselines and fair comparisons: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

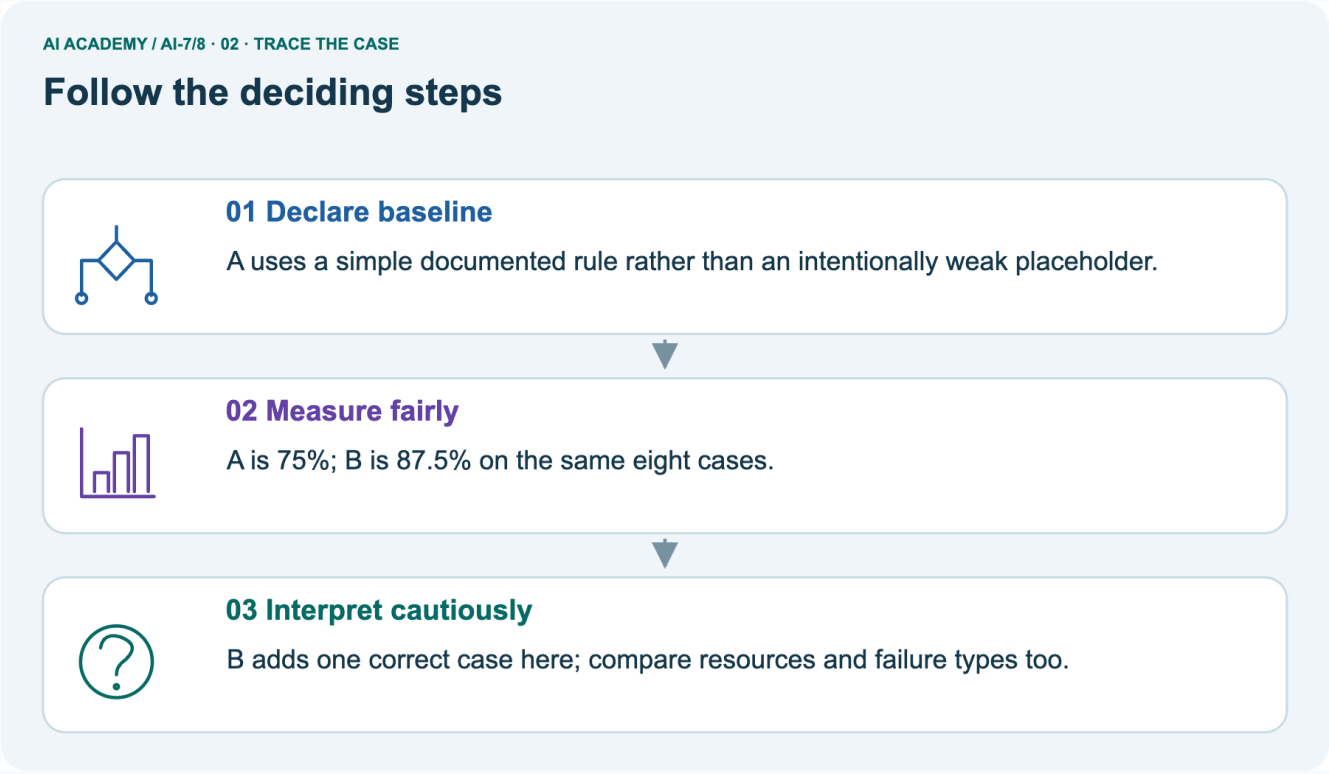
A **baseline** deserves a fair chance. Two classifiers answer the same eight fixed questions with the same allowed sources. Supplied facts: Simple rule A: 6/8 correct. New model B: 7/8 correct.

No test case is used to tune either version.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



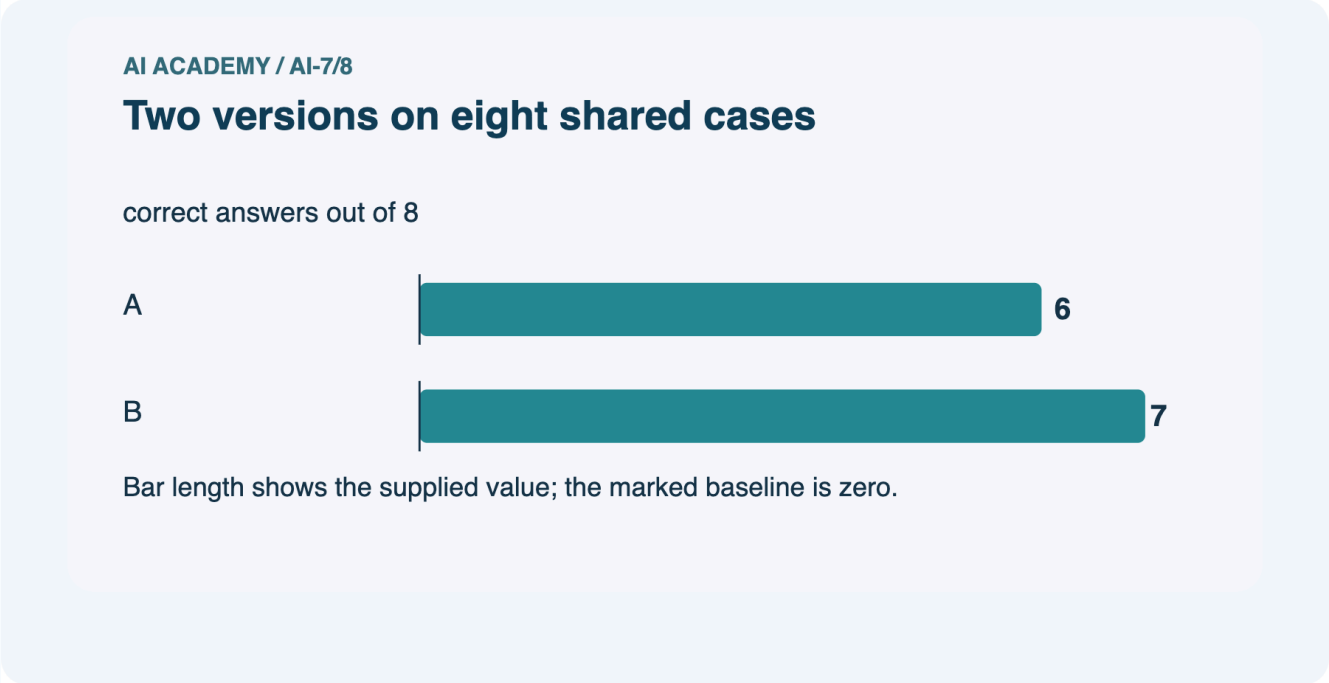
Follow the labelled picture with the matching explanation. Open the vector diagram

Declare **baseline**: A uses a simple documented rule rather than an intentionally weak placeholder. → Measure fairly: A is 75%; B is 87.5% on the same eight cases. → Interpret cautiously: B adds one correct case here; compare resources and failure types too.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Two versions on eight shared cases



Follow the labelled picture with the matching explanation. Open the vector diagram

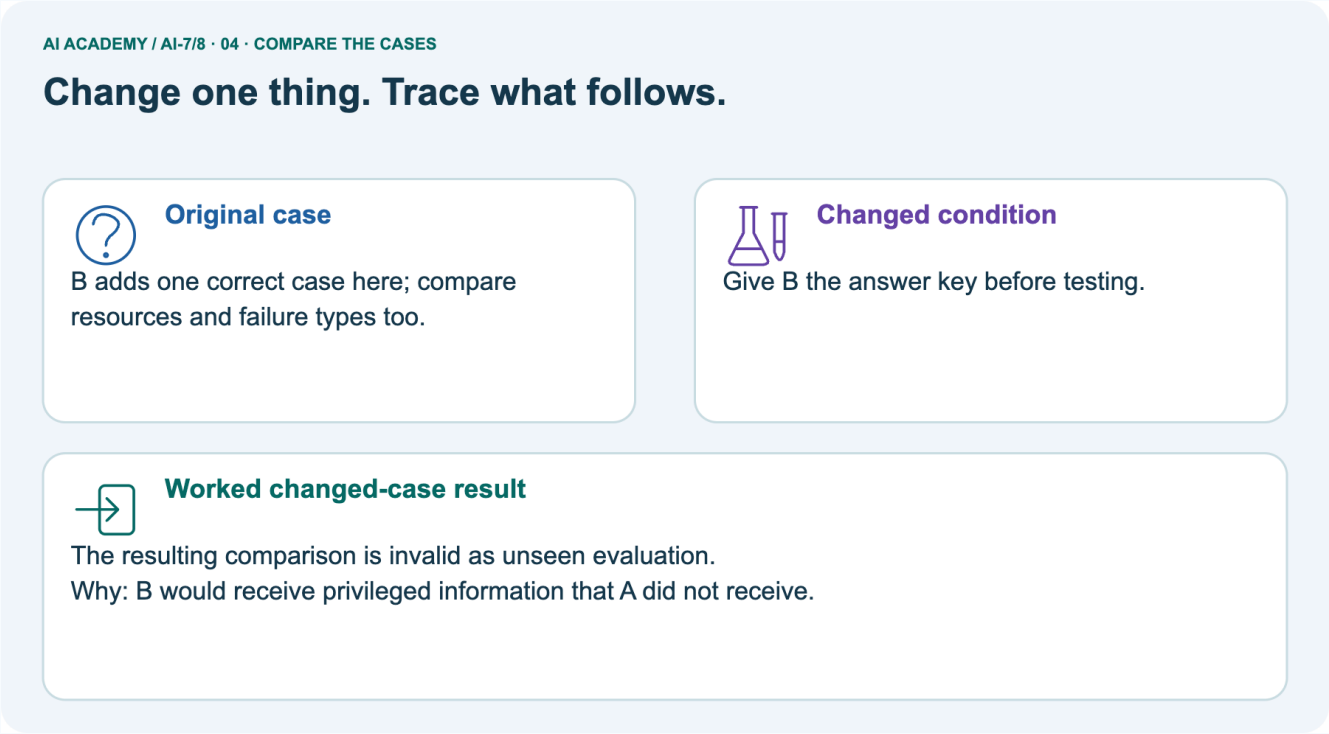
Two versions on eight shared cases. Units: correct answers out of 8. A: 6. B: 7.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: B adds one correct case here; compare resources and failure types too. Changed condition: Give B the answer key before testing. Expected result: The resulting comparison is invalid as unseen evaluation.

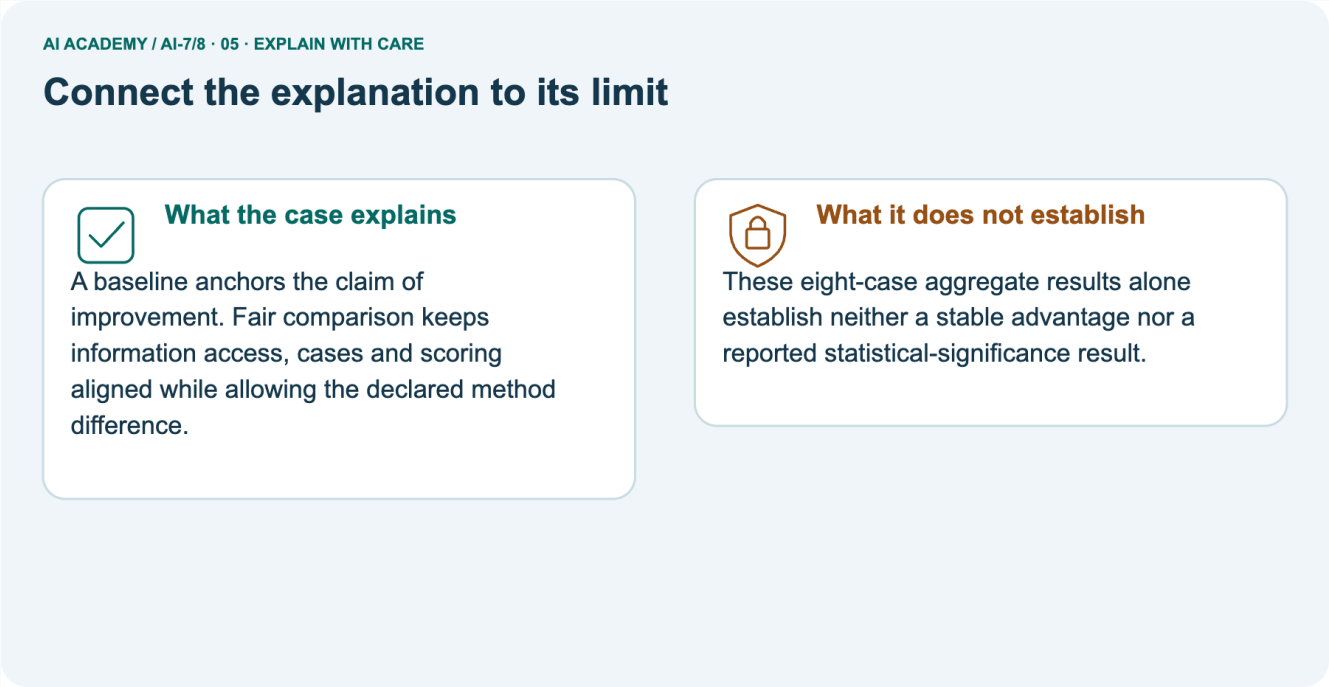
Reason: B would receive privileged information that A did not receive.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: B would receive privileged information that A did not receive.

05. Connect the explanation to its limit



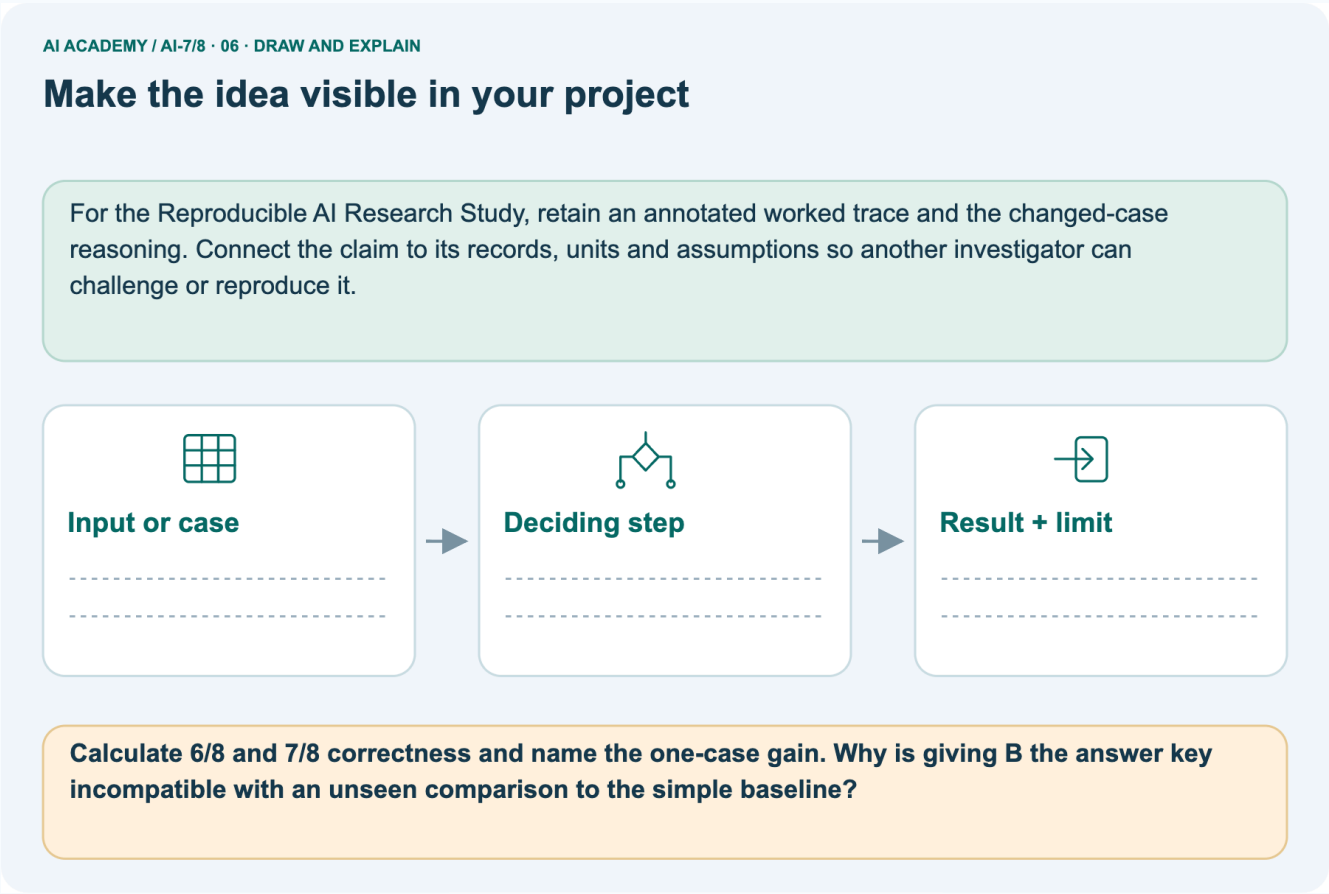
Follow the labelled picture with the matching explanation. Open the vector diagram

A **baseline** anchors the claim of improvement. Fair comparison keeps information access, cases and scoring aligned while allowing the declared method difference. Limit: These eight-case aggregate results alone establish neither a stable advantage nor a reported statistical-significance result.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see. Keep the stated scope: These eight-case aggregate results alone establish neither a stable advantage nor a reported statistical-significance result.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate 6/8 and 7/8 correctness and name the one-case gain. Why is giving B the answer key incompatible with an unseen comparison to the simple **baseline**?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: A uses a simple documented rule rather than an intentionally weak placeholder. A is 75%; B is 87.5% on the same eight cases. B adds one correct case here; compare resources and failure types too.

Changed case: The resulting comparison is invalid as unseen evaluation. B would receive privileged information that A did not receive.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Baselines and fair comparisons: the case

02. Follow the deciding steps

03. Two versions on eight shared cases

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.